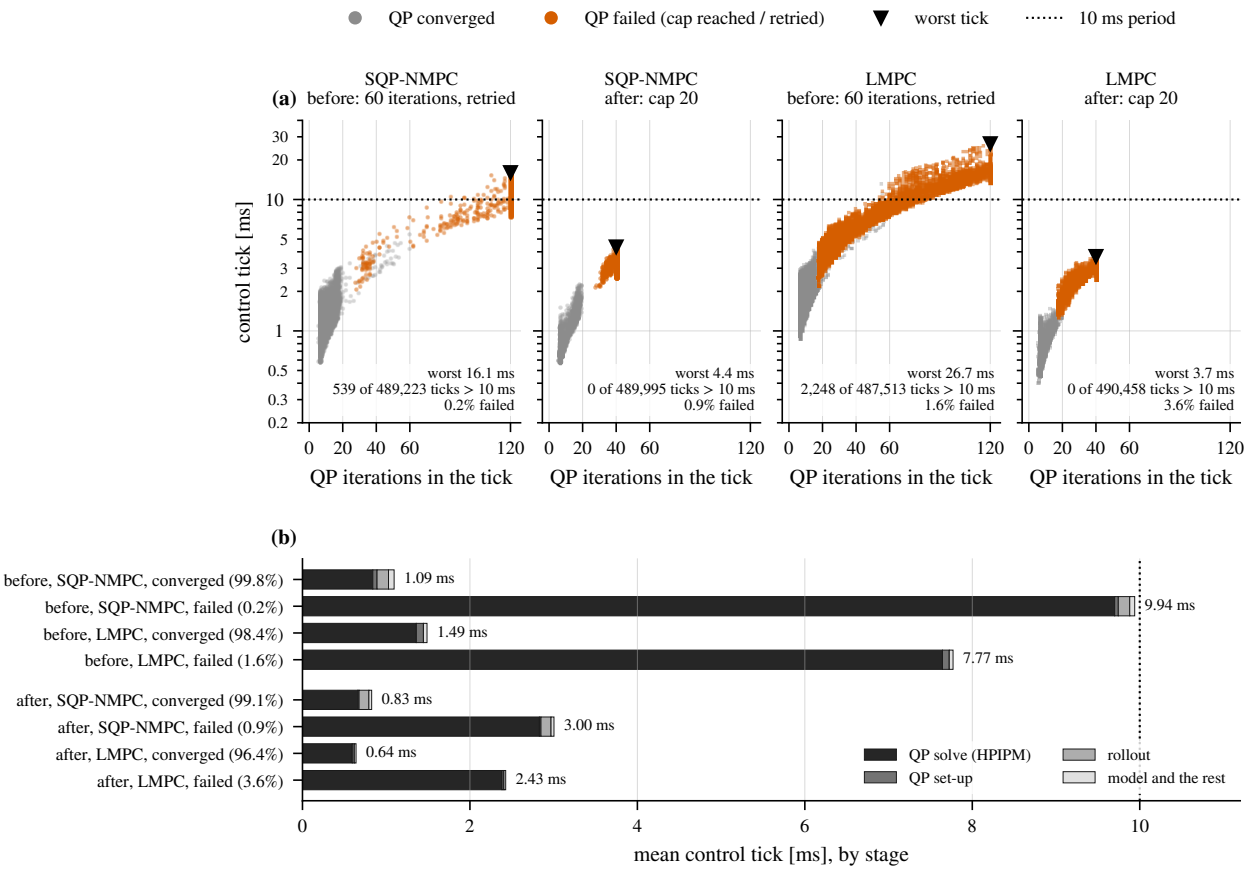
QP converged
QP failed (cap reached / retried)
worst tick
10 ms period
(a)
SQP-NMPC
before: 60 iterations, retried
SQP-NMPC
after: cap 20
LMPC
before: 60 iterations, retried
LMPC
after: cap 20
control tick [ms]
30
20
10
5
3
2
1
0.5
0.3
0.2
worst 16.1 ms
539 of 489,223 ticks > 10 ms
0.2% failed
worst 4.4 ms
0 of 489,995 ticks > 10 ms
0.9% failed
worst 26.7 ms
2,248 of 487,513 ticks > 10 ms
1.6% failed
worst 3.7 ms
0 of 490,458 ticks > 10 ms
3.6% failed
0
20
40
60
120
QP iterations in the tick
(b)
before, SQP-NMPC, converged (99.8%)
before, SQP-NMPC, failed (0.2%)
before, LMPC, converged (98.4%)
before, LMPC, failed (1.6%)
after, SQP-NMPC, converged (99.1%)
after, SQP-NMPC, failed (0.9%)
after, LMPC, converged (96.4%)
after, LMPC, failed (3.6%)
1.09 ms
9.94 ms
1.49 ms
7.77 ms
0.83 ms
3.00 ms
0.64 ms
2.43 ms
QP solve (HPIPM)
QP set-up
rollout
model and the rest
0
2
4
6
8
10
mean control tick [ms], by stage